Venkata Anudeep Adiraju

Data Scientist — Feature Engineering — ETL Pipelines — Statistical & Machine Learning
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PROFESSIONAL SUMMARY

Data Scientist with healthcare analytics expertise, specializing in **feasibility studies**, **feature engineering**, and **ETL pipeline** development for clinical and operational use-cases. Built production systems processing 5M+ health records using **Python, SQL, and Java**, deployed across **AWS, Azure, and GCP** cloud platforms. Expert in applying **statistical and machine learning techniques** to high-dimensionality healthcare data, creating **Grafana dashboards** for business stakeholders, and deriving predictive models that drive clinical and financial outcomes.

TECHNICAL SKILLS

Programming & Data Science: Python (pandas, NumPy, scikit-learn), Java, SQL, R, Statistical Analysis, Predictive Modeling
Machine Learning & Statistics: Statistical Techniques, Model Evaluation, Model Selection, Feature Engineering, A/B Testing, PyTorch
Data Engineering: ETL Pipelines, Data Transformation, Feature Extraction, Data Quality Frameworks, Apache Spark, Batch Processing
Cloud & Infrastructure: AWS (EC2, S3, Lambda, Redshift), Microsoft Azure, Google Cloud (GCP), OCI Cloud, Docker, Kubernetes
Dashboards & Visualization: Grafana, OpenSearch, Oracle Analytics Cloud, Tableau, Plotly, Streamlit, Business Intelligence Reporting
AI/ML Tools: LLMs, RAG Systems, BERT, Vector Databases, MLflow, Model Serving (FastAPI), NLP

EDUCATION

M.S. Computer Engineering | University of Texas at San Antonio | GPA: 3.70/4.0 — Dec 2025
Relevant Coursework: Machine Learning, Statistical Methods, Database Systems, Cloud Computing, Advanced Algorithms
B.Tech Computer Science | HITAM, India | GPA: 3.8/4.0 — Sep 2023

DATA SCIENCE EXPERIENCE

Data Scientist – Healthcare Analytics | University of Texas at San Antonio — **Jan 2024 – Present**

- Conducted **feasibility studies** to assess data availability and quality for clinical research; engineered **feature extraction pipelines** in **Python and SQL** processing 5M+ health records, identifying modeling requirements that enabled 3 new predictive use-cases for business stakeholders
- Designed **ETL pipelines** on **AWS** integrating heterogeneous healthcare data sources (clinical trials, patient records, claims data); built data transformation workflows achieving **42% query performance improvement** supporting downstream statistical analysis
- Applied **statistical and machine learning techniques** to engineer explanatory features from 55K+ healthcare conversation pairs; performed iterative **model selection and evaluation** achieving **82% prediction accuracy** on clinically important treatment outcomes
- Built **Grafana dashboards** and **OpenSearch** visualizations for business teams and clinical leaders; created real-time monitoring tracking system health, **ETL pipeline** latency, and model performance metrics across **500K+ daily requests**
- Developed **Java**-based data ingestion service extracting features from unstructured clinical documents; implemented NLP preprocessing pipeline on **Google Cloud** reducing manual data processing time by **91%** for research teams
- Collaborated with **business stakeholders** to identify data analysis requirements; proposed new uses for existing datasets, delivering predictive models that reduced patient dropout rates by **23%** through early intervention recommendations
- Supported iterative **model experimentation** workflows using MLflow on **Azure**; evaluated 12+ candidate models using cross-validation and statistical significance testing, selecting best-fit algorithms for financially and operationally important healthcare use-cases

Data Science Intern | Aimlytics Technologies — **Mar 2022 – May 2022**

- Built **ETL pipelines** in **Python and SQL** on **Azure** cloud processing behavioral data for **100K+ daily users**; engineered features from user interaction logs enabling predictive churn modeling with **85% accuracy**
- Applied **statistical and machine learning techniques** to engineer features from high-dimensionality user data; delivered insights to **business teams** driving **40% improvement** in API response times through data-driven optimization
- Created **analytics dashboards** tracking A/B test results and user behavior metrics; automated reporting pipelines delivering daily insights to stakeholders enabling data-driven product decisions

DATA SCIENCE PROJECTS

QuitTxt Clinical Data Science Platform | Healthcare Feature Engineering | GitHub

- Led **feasibility study** for multi-institutional smoking cessation research; built **ETL pipelines** in **Python and SQL** processing clinical trial data from UTSA and UT Health San Antonio with HIPAA-compliant data governance
- Engineered features from unstructured health data using **statistical and machine learning techniques** on **AWS**; built RAG pipeline with vector database processing 10K+ medical documents for evidence-based **predictive modeling**
- Developed **Grafana monitoring dashboard** for clinical research teams tracking patient engagement metrics; enabled **business stakeholders** to identify at-risk participants, improving study retention by **18%**

SCRAPE Neural Feature Extraction Engine | Scalable ETL Pipeline | Live Demo

- Architected scalable **ETL pipeline** in **Python** for **feature extraction** from unstructured web data; processed 1K+ URLs/hour using distributed task queue on **GCP** with intelligent rate limiting and error handling
- Built content **feature engineering** workflow with BM25 filtering; achieved **95% extraction accuracy** through iterative **model evaluation**, demonstrating ability to industrialize collection from heterogeneous data sources

CERTIFICATIONS & ACHIEVEMENTS

Cloud Platforms: AWS Certified Cloud Practitioner, Azure Fundamentals (AZ-900), Azure AI Fundamentals (AI-900)
AI/ML: TensorFlow Developer Certificate | **Recognition:** DGX Spark Hackathon Winner – GPU-accelerated ML pipelines